

# JaeMin Bok

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## Technical Skills

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**Programming:** Python, JavaScript, SQL (PostgreSQL, MySQL, MS SQL Server), C  
**Frontend & Web:** React, FastAPI, Streamlit, REST API  
**AI / ML & Tools:** LLM & RAG application integration, LangChain, Unity, Figma, Notion, GitHub  
**Languages:** Korean (Native), English (Intermediate)

## Experience

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| <b>Software Engineering Internship</b><br><i>Smart Quality Laboratory</i>                                                                                                                                                                                                            | Mar. 2025 – Feb. 2026<br><i>Suwon, South Korea</i> |
| <ul style="list-style-type: none"><li>– Built a digital twin using Unity to achieve space quality in a plant environment.</li><li>– Developed a color leadership game UI using Unity.</li><li>– Created an LLM-based manual assistant interface designed for senior users.</li></ul> |                                                    |
| <b>Undergraduate Research Student</b><br><i>e-Manufacturing Laboratory</i>                                                                                                                                                                                                           | Mar. 2026 – Present<br><i>Suwon, South Korea</i>   |
| <ul style="list-style-type: none"><li>– Studied Digital Twin technologies and their applications in manufacturing.</li><li>– Researched optimization algorithms for production scheduling problems.</li></ul>                                                                        |                                                    |

## Selected Projects

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**NeuroFactory: Ontology-Grounded Hybrid RAG for Natural Language Querying in MES** *Capstone Design Project*  
*Role: Frontend Developer*

- Built the React-based frontend UI for an on-premises NL-to-SQL system, letting factory engineers submit Korean/English natural-language queries over a live MES (THiRA MES 3.1.0) and view results without SQL knowledge.
- Implemented query input, tabular result visualization, validation-error feedback, and diagnostics views in the user interface.
- Integrated the frontend with the FastAPI backend (REST API), wiring the end-to-end query pipeline (retrieval → SQL generation → safety validation → execution) to the UI.
- Contributed to a system achieving **89.5% execution accuracy** and **100% valid-SQL rate** on the MES-NL2SQL-Gold benchmark (n=200), a **+52.5 pp** gain over the schema-only baseline.

**Automatic Semiconductor Fault Diagnosis**  
*Role: Developer*

- Designed an automatic semiconductor fault diagnosis system using computer vision methods and Streamlit.
- Applied image preprocessing and feature extraction techniques to improve defect detection accuracy.
- Built a mapping framework to trace classified defects back to corresponding faulty manufacturing processes.
- Implemented an interactive interface using Streamlit for real-time visualization and analysis.

## Research & Achievements

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**Patent (Filed):** “Method and System for Role-Based Natural Language Query Processing in Manufacturing Execution Systems” 2026  
*Korean Patent Application No. R-2026-0542-KR-1 (filed Jun. 2026)*

**Software Copyright Registration:** “NeuroFactory — Natural Language Query Engine for Manufacturing Execution Systems” 2026  
*Korea Copyright Commission, Reg. No. C-2026-030625*

## Publications

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B. Jung, W. Kim, G. Suh, **J. Bok**, and J. Jeong, “NeuroFactory: Ontology-Grounded Hybrid RAG for Natural Language Querying in Manufacturing Execution Systems,” in *Proc. IEEE Int. Conf. Big Data (BigData) Workshops, BD-SMDT*, 2026. (under review)

## Honors & Awards

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**ADSP** (Advanced Data Analytics Semi-Professional) 2024

## Education

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| <b>Sungkyunkwan University (SKKU)</b><br><i>B.S. in System Management Engineering and Software Engineering</i>                                    | Mar. 2023 – Present<br><i>Suwon, South Korea</i> |
| <ul style="list-style-type: none"><li>– Relevant Coursework: Data Structures, Algorithms, Databases, Computer Systems, Machine Learning</li></ul> |                                                  |